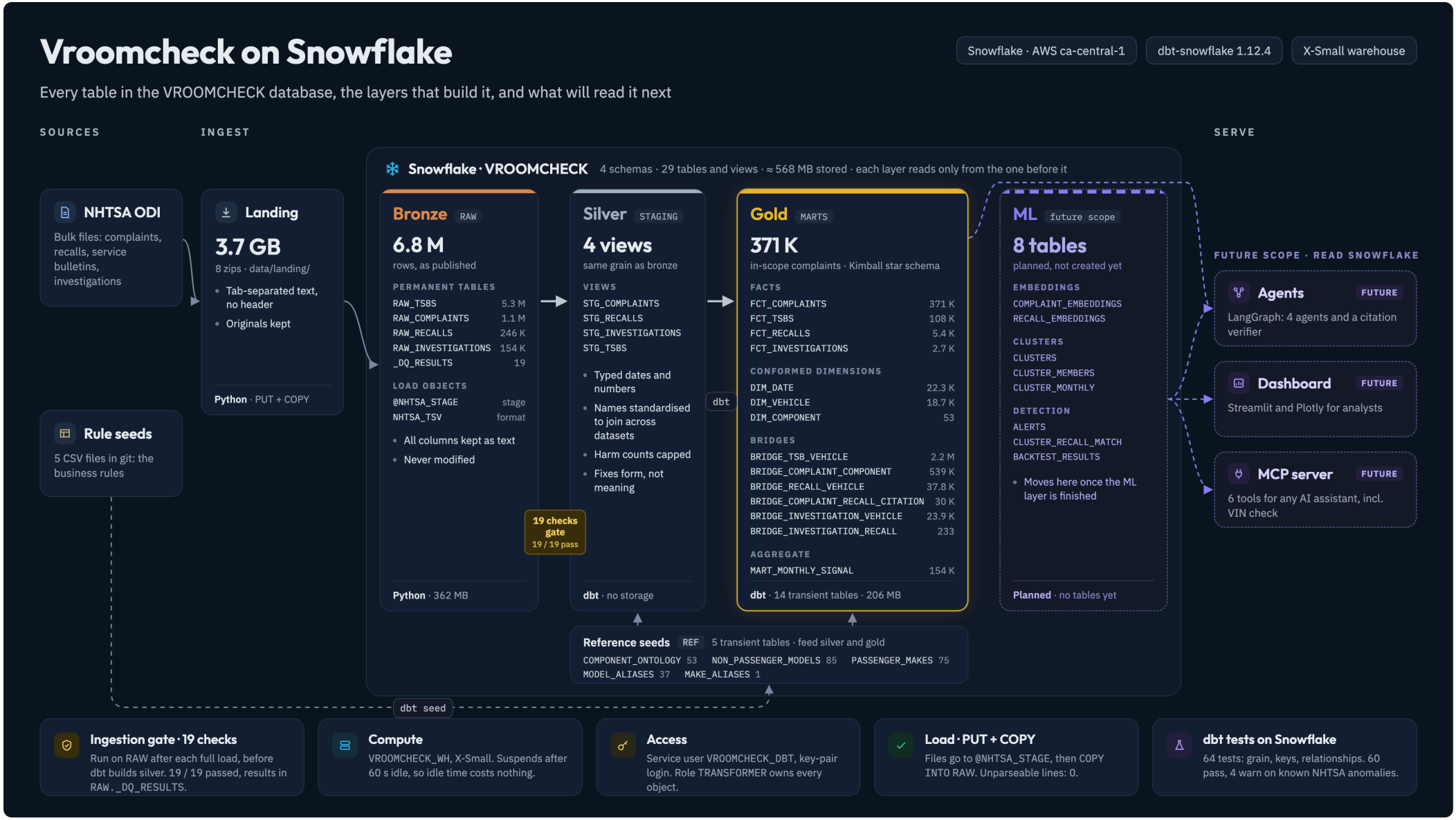


The slide these notes go with: Vroomcheck on Snowflake (docs/Vroomcheck_Snowflake_Warehouse_Dark.png).

Snowflake warehouse diagram · presentation notes

Vroomcheck · Part 1 is the detailed explanation with likely questions · Part 2 is a script to read in 4–5 minutes

Part 1 · Detailed explanation

Follows the diagram left to right, then the bottom row

Opening

Point at the title

- This is our data warehouse on Snowflake: one database called **VROOMCHECK**, with 4 schemas and 29 tables and views.
- It follows the **medallion pattern**: bronze, silver, gold. Each layer reads only from the one before it, so every number in gold can be traced back to an original NHTSA record.

1 · Sources and ingest

Point at the far left

- The data comes from **NHTSA's public bulk files**: owner complaints, recalls, investigations and technical service bulletins.
- We download **8 zip files, about 3.7 GB** of tab-separated text, and keep the originals untouched.
- A Python loader uploads them to a Snowflake **internal stage** with `PUT`, then loads them with `COPY INTO`. That's Snowflake's standard bulk-load path.
- The **rule seeds** at the bottom left are our business rules, kept as small CSV files in git.

2 · Bronze: RAW

Point at the orange column

- Bronze is the data **exactly as NHTSA published it: 6.8 million rows**, every column stored as text.
- We **never modify** these tables. If anything goes wrong downstream, we can always rebuild from here.
- These are **permanent tables**, so they keep Snowflake's Time Travel and Fail-safe protection. It's the only copy of the source data, so it's worth protecting.
- The column also shows the stage and the file format we load through, and `_DQ_RESULTS`, where the quality checks write their results.

3 · The 19-check gate

Point at the yellow badge between bronze and silver

- Before anything moves to silver, the data goes through a **gate of 19 quality checks**.
- For each dataset they check that the table isn't empty, the IDs aren't blank, the dates parse, and no lines were rejected. There are also checks for missing complaint text, complaints dated in the future, and badly formatted recall numbers.
- **12 of the 19 are critical**: if any of them fails, the pipeline stops and dbt never runs. **All 19 pass** today, and every run is recorded in `RAW._DQ_RESULTS`, so we have a history of data health.

4 · Silver: STAGING

Point at the grey column

- Silver is **4 views** built by dbt. The rule here is **"fix form, not meaning"**.
- They turn text into real dates and numbers, standardise make and model names so the same car matches across all four datasets, and cap obviously impossible harm counts.
- They're **views, so they take no storage**. Snowflake runs them when they're queried, so they're always current.

5 · Gold: MARTS

Point at the highlighted yellow column

- Gold is where the analysis happens: a **Kimball star schema** with **371,000 in-scope complaints**.
- **4 fact tables**, one per business event: complaints, recalls, investigations and service bulletins.
- They share **3 conformed dimensions**: vehicle, date and component. The vehicle dimension is what lets us compare complaints, recalls and investigations for the same car.
- **6 bridge tables** handle the many-to-many links, like one recall covering many vehicles. `MART_MONTHLY_SIGNAL` is the monthly summary our detection uses.
- These are **14 transient tables, 206 MB**, rebuilt by dbt on every run. Because they can always be rebuilt from bronze, we skip Fail-safe and save on storage.

6 · Reference seeds

Point at the card under silver and gold

- These are the **business rules as data**: the component ontology with severity weights, the passenger-vehicle make list, and the name aliases.
- To change a rule, we edit one row in a CSV, not the SQL. dbt loads them with `dbt seed`, and they feed both silver and gold.

7 · ML and future scope

Point at the dashed purple parts on the right

- The dashed parts are **future scope**. The ML column shows the **8 tables we plan to add**: embeddings, clusters and detection results.
- Once they're here, our **LangGraph agents, the Streamlit dashboard and the MCP server** can read straight from Snowflake. That's the next phase.

8 · Bottom row

One sentence each

- **Ingestion gate:** 19 checks on RAW after every load, before dbt.
- **Compute:** an **X-Small warehouse that suspends after 60 seconds idle**, so we only pay while queries run. Storage and compute are separate in Snowflake.
- **Access:** a dedicated **service user with key-pair login** and one role, TRANSFORMER, that owns everything. No passwords, least privilege.
- **Load:** PUT and COPY INTO, with **zero unparseable lines**.
- **dbt tests: 64 automated tests** on grain, keys and relationships: 60 pass, and 4 warn on known quirks in NHTSA's data.

Closing line

> *“So: raw data lands in bronze untouched, a 19-check gate guards it, dbt cleans it into silver and models it into a Kimball star schema in gold, all inside Snowflake, tested, and costing almost nothing when idle. The dashed parts show where the ML layer and the apps plug in next.”*

Questions you might get

Why views for silver but tables for gold?
Views cost nothing to store and are always current. Gold is what gets queried heavily, so it's stored as tables for speed.

Why transient tables?
Gold and the seeds can be rebuilt from bronze in about 20 seconds, so Fail-safe backup storage would be wasted money. Bronze is the only copy of the source, so it stays permanent.

What happens if a check fails?
If it's one of the 12 critical checks, the load stops before dbt runs, so bad data never reaches silver or gold. A warning check is logged, and the run continues.

Why isn't the ML layer in Snowflake yet?
We're still developing it. We wanted the core warehouse stable and tested first.

How much does it cost?
An X-Small warehouse uses 1 credit an hour, but only while it runs. A full dbt build takes about 20 seconds, and the warehouse suspends after a minute idle.

What's the difference between `_LOAD_REJECTS` and the gate?
`_LOAD_REJECTS` catches lines Snowflake couldn't parse during the load (there are 0). The gate then checks whether the loaded data makes sense.

Part 2 · Speaking script

About 640 words, roughly 4.5 minutes at a comfortable pace

Section	Time
Introduction	0:30
Sources and ingestion	0:30
Bronze layer	0:40
The quality gate	0:40
Silver layer	0:30
Gold layer	0:50
Reference seeds	0:20
Future scope	0:30
Platform and conclusion	0:40
Total	**≈ 4:30**

Introduction 0:30

Point at the title

This diagram shows our data warehouse on Snowflake. Everything lives in one database called VROOMCHECK, which has four schemas and twenty-nine tables and views. We follow the medallion pattern of bronze, silver and gold, and each layer reads only from the one before it. That gives us full lineage: any number in the final star schema can be traced back to an original NHTSA record.

Sources and ingestion 0:30

Point at the left side

Our data comes from NHTSA's public bulk files: owner complaints, recalls, defect investigations and technical service bulletins. We download eight zip files, about 3.7 gigabytes of tab-separated text, and we keep the originals untouched. A Python loader uploads these files to a Snowflake internal stage using the PUT command, then loads them into tables with COPY INTO, which is Snowflake's standard bulk-loading method. Below that are our rule seeds, small CSV files in git that hold our business rules. I'll come back to those in a moment.

Bronze layer 0:40

Point at the orange Bronze column

Bronze is the raw layer. It holds the data exactly as NHTSA published it: about 6.8 million rows, with every column stored as text. We never modify these tables, so if anything goes wrong further down, we can always rebuild from here. Because this is the only copy of our source data, these are permanent tables, which means they keep Snowflake's Time Travel and Fail-safe protection. You can also see the stage and file format we load through, and the results table for our quality checks.

The quality gate 0:40

Point at the yellow badge between Bronze and Silver

Before any data moves to silver, it has to pass a gate of nineteen quality checks. For each dataset, we check that the table is not empty, that the IDs are not blank, that the dates can be parsed, and that no lines were rejected during loading. We also check for missing complaint text, complaints dated in the future, and badly formatted recall numbers. Twelve of these checks are critical, so if any of them fails, the pipeline stops and dbt never runs. Today, all nineteen checks pass, and every run is saved in Snowflake so we keep a history of data health.

Silver layer 0:30

Point at the grey Silver column

Silver is built by dbt and consists of four views. Our rule here is to fix the form of the data, not its meaning. These views turn text into proper dates and numbers, standardise make and model names so the same car matches across all four datasets, and cap harm counts that are clearly impossible. Because they are views, they use no storage and are always up to date.

Gold layer 0:50

Point at the highlighted Gold column

Gold is where the analysis happens, and it is designed as a Kimball star schema with about 371,000 in-scope complaints. There are four fact tables, one for each business event: complaints, recalls, investigations and service bulletins. They share three conformed dimensions: vehicle, date and component. The shared vehicle dimension is what allows us to compare complaints, recalls and investigations for exactly the same car. Six bridge tables handle many-to-many relationships, such as one recall covering many vehicles. There is also a monthly signal table that our detection logic uses. These fourteen tables are transient, because dbt can rebuild them from bronze at any time, so we do not pay for extra backup storage.

Reference seeds 0:20

Point at the card under Silver and Gold

These seed tables hold our business rules as data: the component ontology with severity weights, the list of passenger vehicle makes, and the name aliases. If a rule changes, we edit one row in a CSV file instead of rewriting SQL.

Future scope 0:30

Point at the dashed purple parts on the right

The dashed parts show our future scope. The ML column lists the eight tables we plan to add for embeddings, clusters and detection results. Once those are in Snowflake, our LangGraph agents, our Streamlit dashboard and our MCP server will be able to read directly from the warehouse.

Platform and conclusion

0:40

Point at the bottom row

Along the bottom are the platform details. For compute, we use an X-Small warehouse that suspends after sixty seconds of inactivity, so we only pay while queries are running. For access, a dedicated service user logs in with a secure key pair, and one role owns every object. And sixty-four automated dbt tests check grain, keys and relationships on every build.

To summarise: raw data lands in bronze untouched, a nineteen-check gate protects it, and dbt cleans it into silver and models it into a Kimball star schema in gold, all inside Snowflake. Thank you.

Timing tips

- **Running short of time:** shorten Reference seeds and Future scope to one sentence each. That saves about 30 seconds.
- **Extra time:** add one example in the Gold section, e.g. “for example, we can count every complaint, recall and investigation for a 2019 Honda Civic with a single join on the vehicle key.”